

MonoTAKD: Teaching Assistant Knowledge Distillation for Monocular 3D Object Detection

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● Problem/Objective

- Monocular 3D object detection
- Knowledge Distillation

● Contribution/Key Idea

- TA model을 이용한 distillation
- 3D spatial cues를 residual feature로 이용
 - teacher $\longleftrightarrow$ TA difference를 capture에 사용
- KITTI 3D 에서 sota

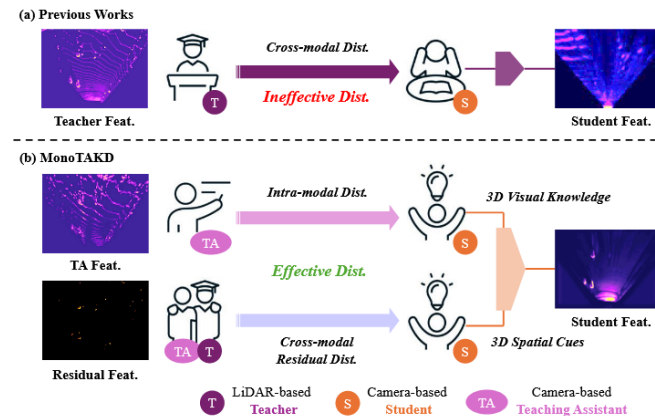
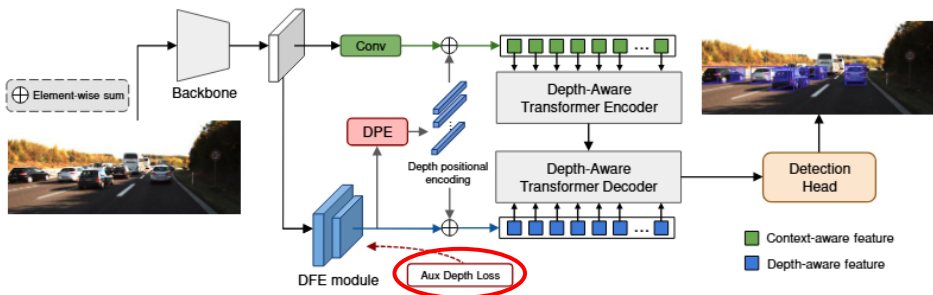
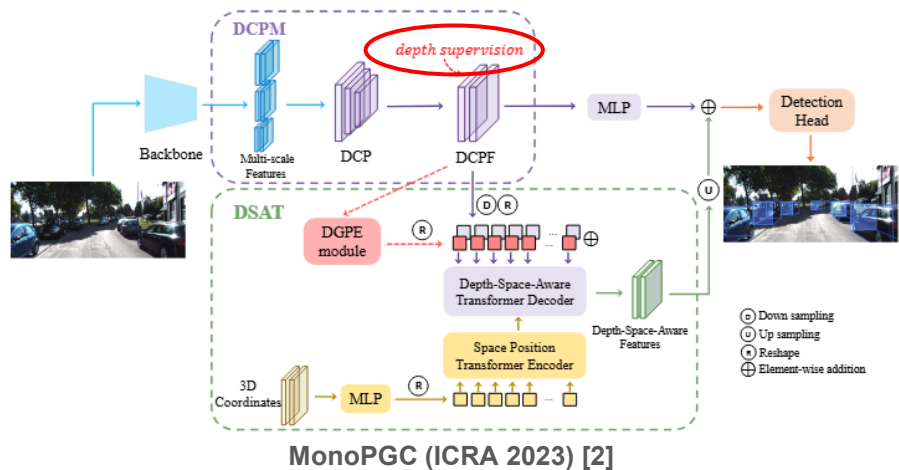


Figure 1. Comparison between previous methods and MonoTAKD (Ours). Previous works [7, 12] face a significant challenge in the distillation process due to a substantial gap in feature representation. Our MonoTAKD incorporates intra-modal and cross-modal residual distillation to enhance learning across this feature representation gap. We visualize the BEV features of each model, along with the residual features derived between the TA and T models, highlighted in orange. Best view with zoom-in and color.

● Introduction



MonoDTR (CVPR 2022) [1]



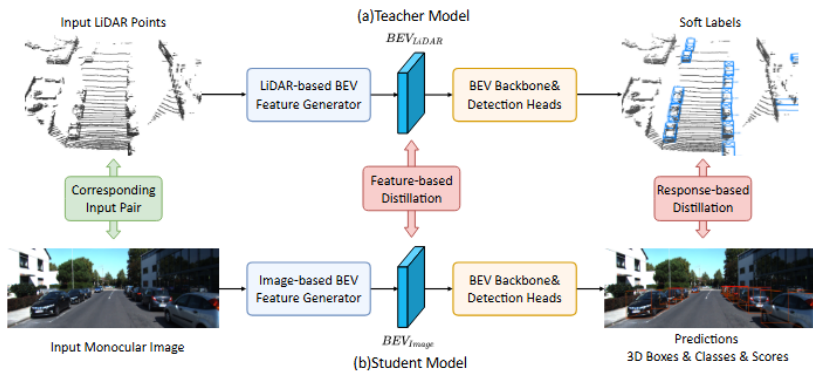
MonoPGC (ICRA 2023) [2]

- Monocular 3D object detection
 - 장점: lightweight
 - 단점: single image → depth estimation Hard
 - Auxiliary supervision을 통한 depth 학습

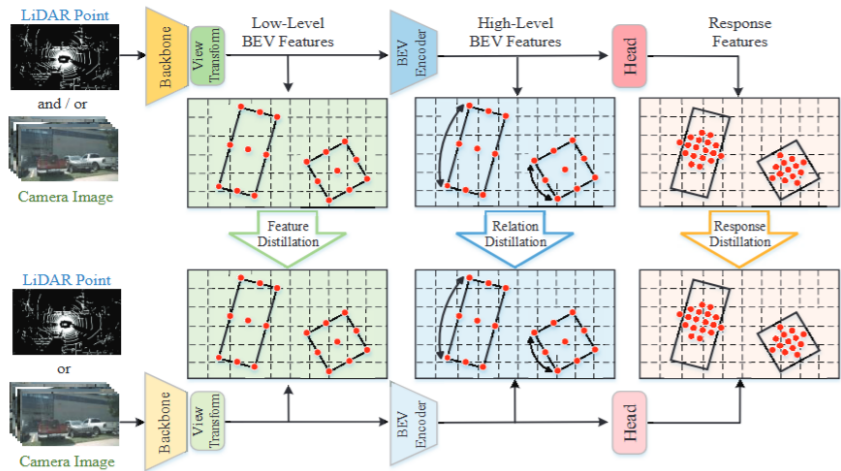
[1] Huang, Kuan-Chih, et al. "Monodtr: Monocular 3d object detection with depth-aware transformer." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022

[2] Wu, Zizhang, et al. "Monopgc: Monocular 3d object detection with pixel geometry contexts." arXiv preprint arXiv:2302.10549 (2023).

● Introduction



CMKD (ECCV 2022) [3]



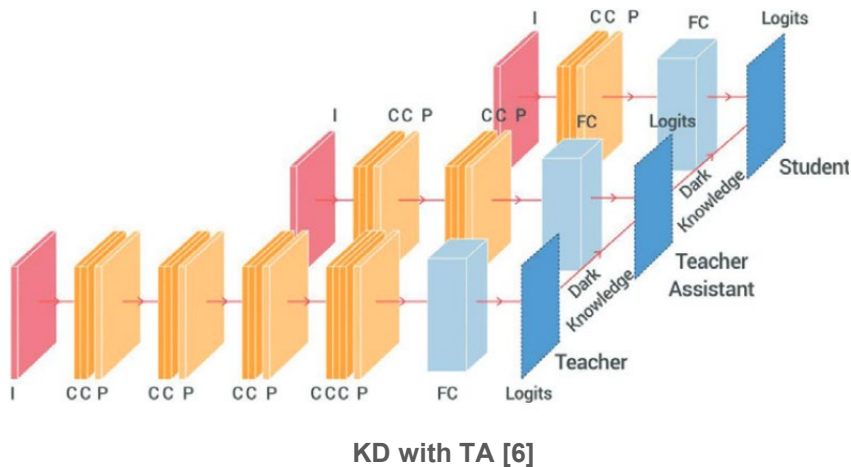
UniDistill (CVPR 2023) [4]

- To Enhance 3D info
 - Cross-modal (L → C) 3D info transfer
 - Auxiliary supervision을 통한 depth 학습

[3] Hong, Yu, Hang Dai, and Yong Ding. "Cross-modality knowledge distillation network for monocular 3d object detection." European Conference on Computer Vision. Cham: Springer Nature Switzerland, 2022.

[4] Zhou, Shengchao, et al. "Unidistill: A universal cross-modality knowledge distillation framework for 3D object detection in bird's-eye view." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2023.

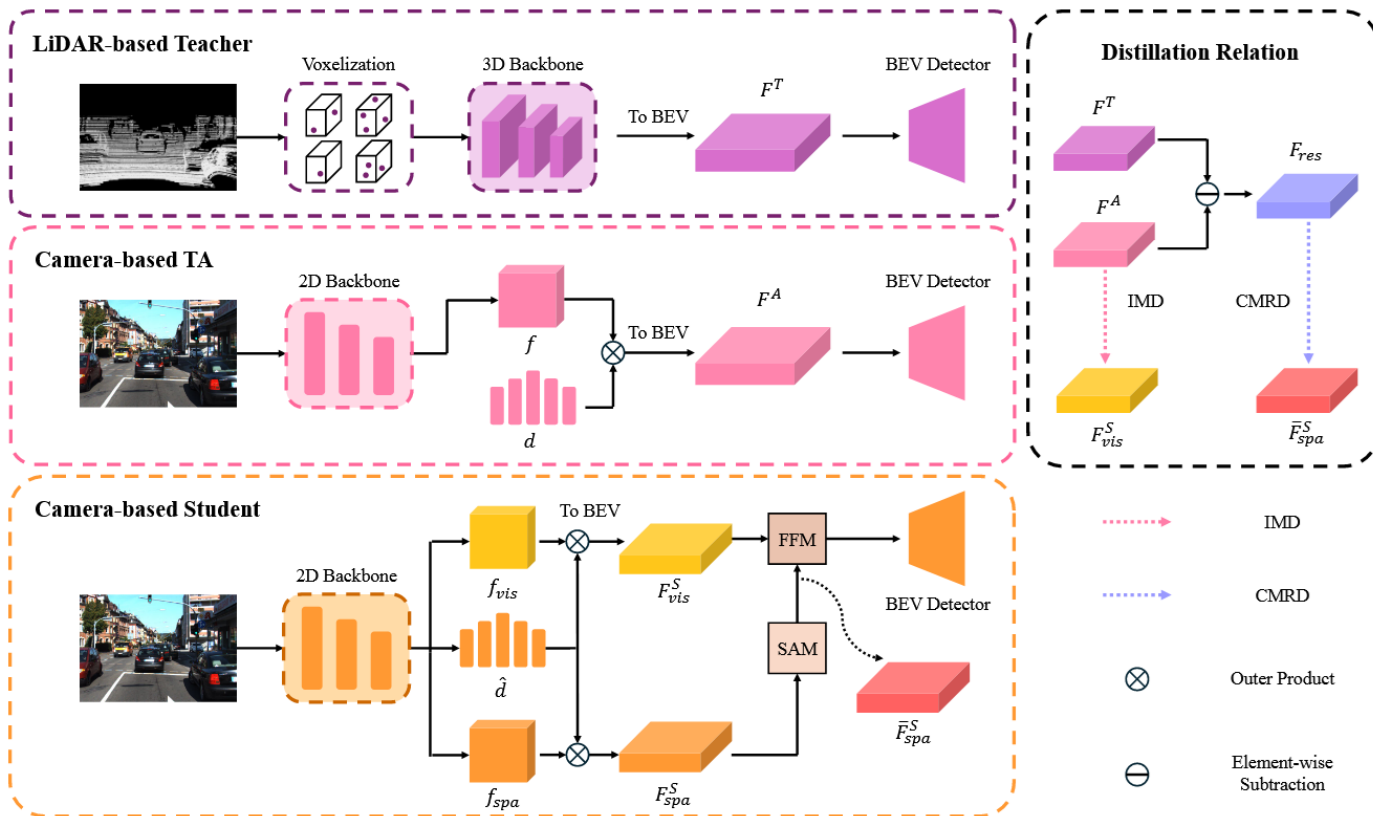
● Introduction



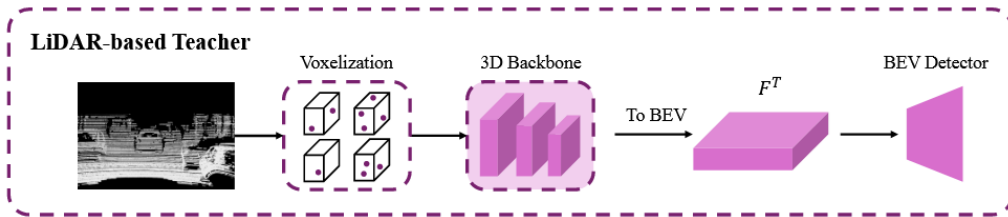
- 그 Modality Gap을 TA를 통해서 해결하겠다.
 - multi-step이 아닌 **End-to-End** 로!

Additionally, we use end-to-end rather than multi-step [26] distillation during training.

Method

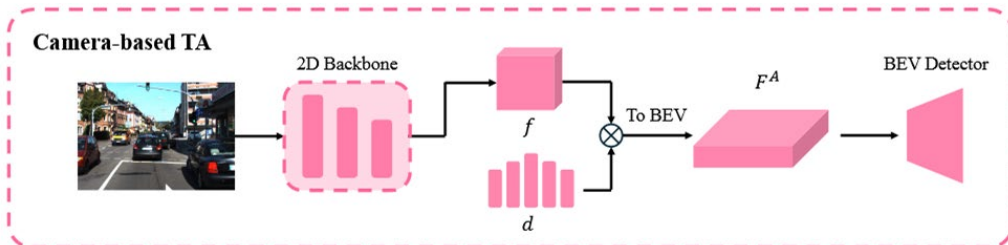


● Method



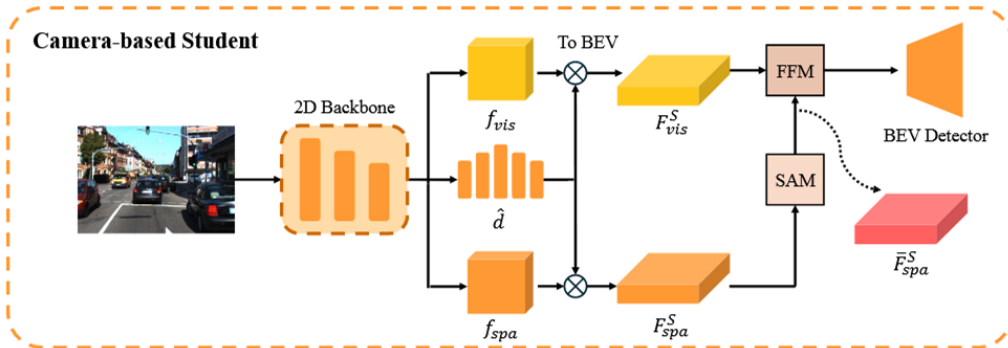
$$F^T \in \mathbb{R}^{H \times W \times C}$$

$$F^T = \phi(\psi(v))$$



ground truth (GT) depth maps $d \in \mathbb{R}^{H \times W \times D}$

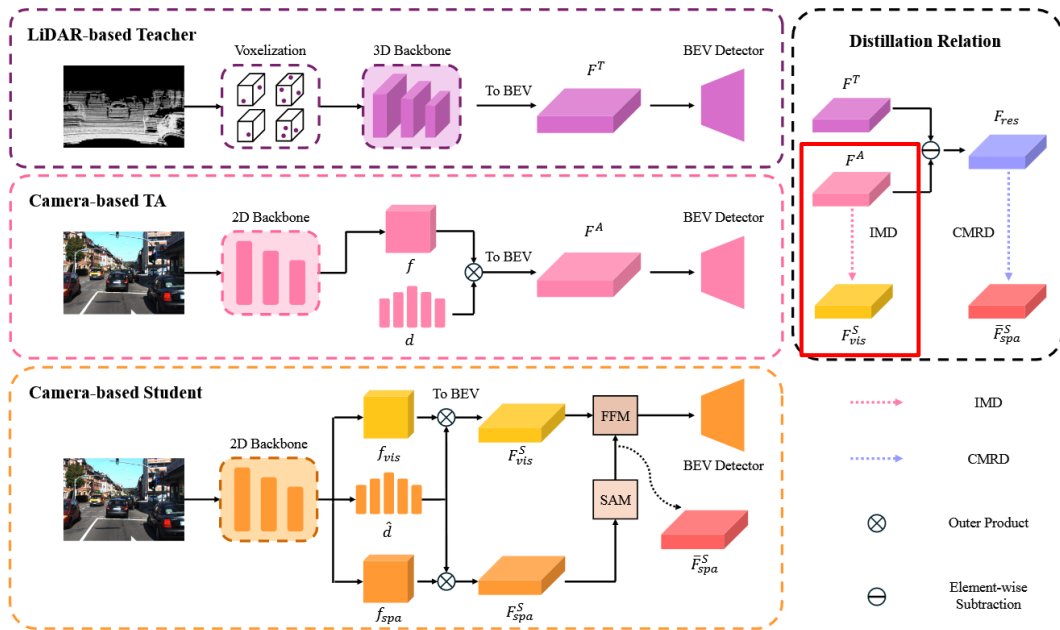
$$F^A = \phi(f \otimes d)$$



f_{vis} : TA로 부터 distill branch

f_{spa} : Teacher-TA 관계를 distill 받는 branch

● Method



$$\mathcal{L}_{IMD} = \text{MSE}(F_{vis}^S, F^A)$$

- 정확한 3D info (depth)를 TA로부터 학습
 - Model 구조도 동일, $S \rightarrow TA$ 화
- 그림에도 TA도 완벽하지 않은 모듈로 LiDAR Teacher로부터 배울점 존재

Method

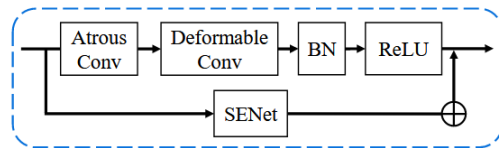
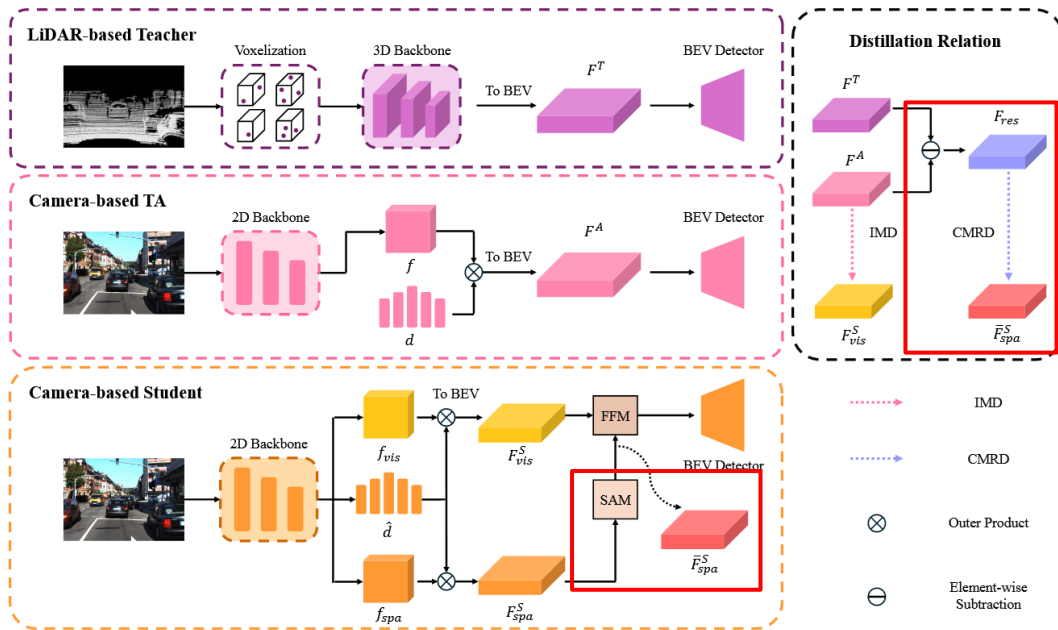


Figure 3. **Spatial Alignment Module (SAM)**. SAM cascades the Atrous and Deformable convolutions to learn the alignment of BEV features. SENet is adopted for channel attention.

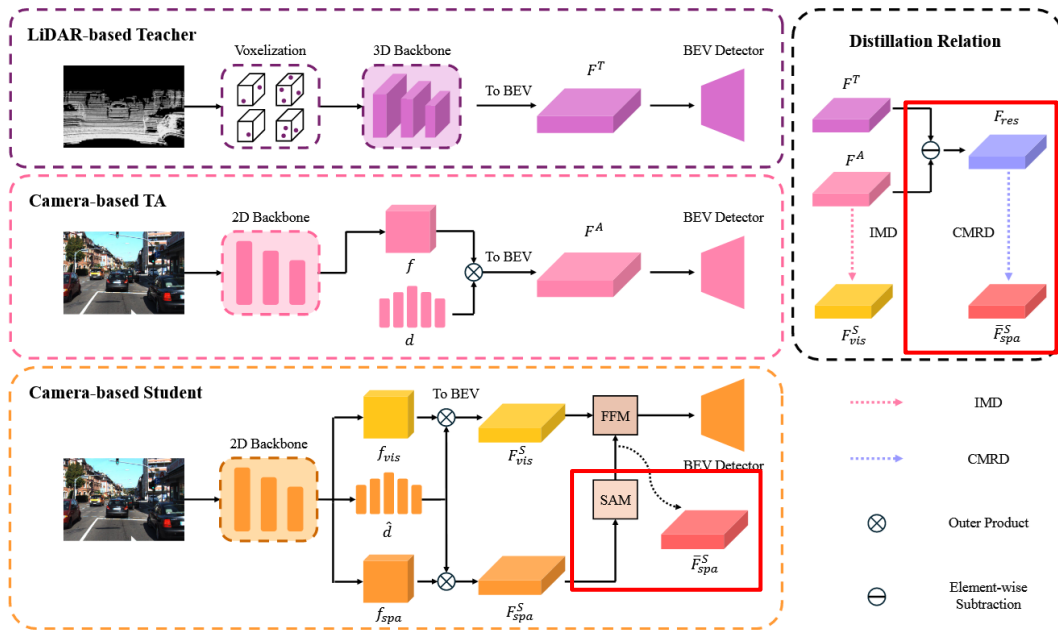
$$\mathcal{L}_{CMRD} = \text{MSE}(\bar{F}_{spa}^S, F_{res})$$

$$F_{res} = F^T \ominus F^A$$

- S → TA화 되었다고 해도 LiDAR만이 가지는 exclusive한 정보는 x
 - TA (GT depth) ↔ Teacher (LiDAR) 차이가 그 exclusive한 정보를 의미

- 그럼 Teacher - TA를 해서 그 residual을 distill 하자

Method



$$\mathcal{L}_{CMRD} = \text{MSE}(\bar{F}_{spa}^S, F_{res})$$

$$F_{res} = F^T \ominus F^A$$

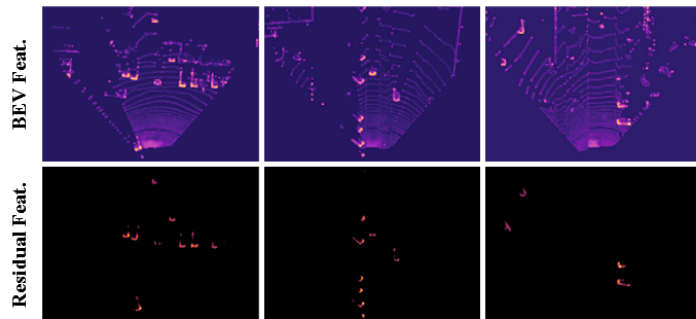
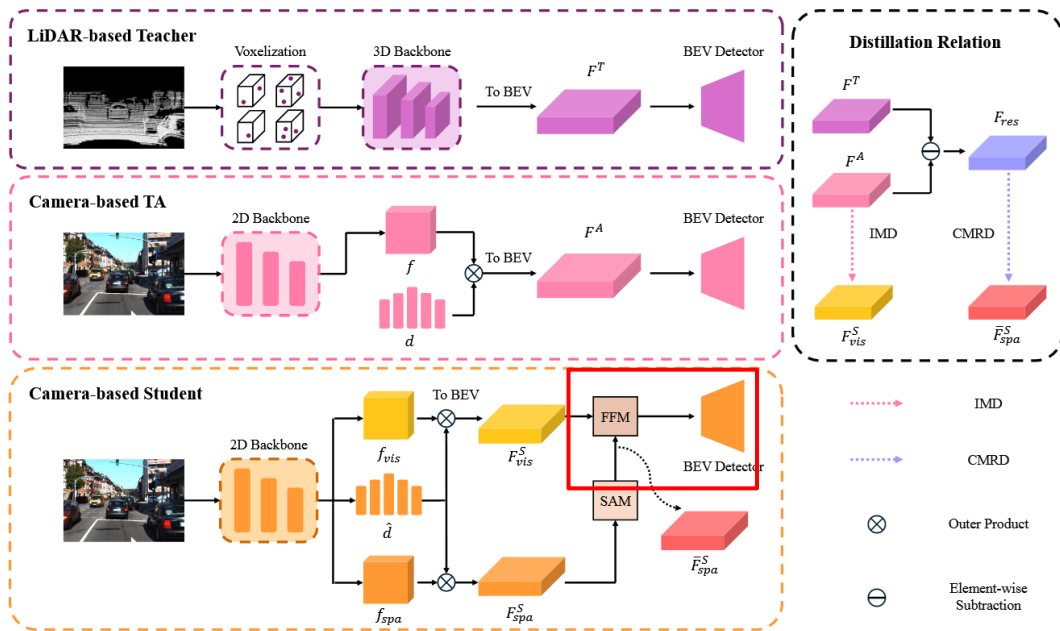


Figure 4. Comparing the BEV features (F^T) from the teacher model with the residual features (F_{res}) reveals that our residuals effectively capture essential 3D spatial cues, emphasizing critical information over less important elements, such as ripples and background noise present in F^T .

- 그냥 Teacher와 달리 Residual은 중요한 정보만 남음

• Method



$$F^S = \text{FFM}(F_{vis}^S \oplus \bar{F}_{spa}^S)$$

- element-wise addition

$$\mathcal{L}_{total} = \mathcal{L}_{IMD} + \mathcal{L}_{CMRD} + \mathcal{L}_{logit}$$

$$\mathcal{L}_{logit} = \mathcal{L}_{cls} + \mathcal{L}_{reg}$$

Experiments

Table 1. Experimental results on the KITTI *test* set for the Car category. We use **bold** and underline to indicate the best and the second-best results, respectively. † denotes the KD-based methods.

Method	Venue	Extra Data	AP_{3D}			AP_{BEV}		
			Easy	Mod.	Hard	Easy	Mod.	Hard
DDMP-3D [36]	CVPR 21	Pre-trained Depth	19.71	12.78	9.80	28.08	17.89	13.44
DD3D [27]	ICCV 21		23.22	16.34	14.20	30.98	22.56	20.03
Kinematic3D [1]	ECCV 20	Temporal	19.07	12.72	9.17	26.69	17.52	13.10
Dfm [1]	ECCV 22		22.94	16.82	14.65	31.71	22.89	19.97
CaDDN [32]	CVPR 21	LiDAR Auxiliary	19.17	13.41	11.46	27.94	18.91	17.19
MonoDTR [14]	CVPR 22		21.99	15.39	12.73	28.59	20.38	17.14
MonoNerd [43]	ICCV 23		22.75	17.13	15.63	31.13	23.46	20.97
MonoPGC [41]	ICRA 23		24.68	17.17	14.14	32.50	23.14	20.30
OccupancyM3D [30]	CVPR 24		25.55	17.02	14.79	35.38	24.18	21.37
MonoDistill† [7]	ICLR 22	None	22.97	16.03	13.60	31.87	22.59	19.72
Cube R-CNN [2]	CVPR 23		23.59	15.01	12.56	31.70	21.20	18.43
MonoUNI [17]	NeurIPS 23		24.75	16.73	13.49	33.28	23.05	19.39
MonoATT [51]	CVPR 23		24.72	<u>17.37</u>	15.00	<u>36.87</u>	<u>24.42</u>	<u>21.88</u>
MonoDETR [49]	ICCV 23		25.00	16.47	13.58	33.60	22.11	18.60
CMKD† [12]	ECCV 22		25.09	16.99	<u>15.30</u>	33.69	23.10	20.67
ADD† [42]	AAAI 23		<u>25.61</u>	16.81	13.79	35.20	23.58	20.08
MonoCD [44]	CVPR 24		25.53	16.59	14.53	33.41	22.81	19.57
MonoTAKD	-	None	27.91	19.43	16.51	38.75	27.76	24.14

• Experiments

Table 2. Experimental results on the nuScenes *val* set. We use **bold** to indicate the best results in each setting.

Method	Modality	Backbone	NDS↑	mAP↑
PETrv2 [25]	C	R50	0.456	0.349
P2D [18]	C	R50	0.474	0.360
PGD [38]	C	R101	0.428	0.369
MonoDETR [49]	C	R101	0.526	0.423
BEVFormer [23]	C	R50	0.423	0.352
+BEVDistill [6]	$L \rightarrow C$	R50	0.457	0.386
+DistillBEV [39]	$L \rightarrow C$	R50	0.476	0.367
+STXD [16]	$L \rightarrow C$	R50	0.481	0.374
+TAKD (Ours)	$L \rightarrow C$	R50	0.490	0.392
BEVFormer	C	R101	0.445	0.374
+BEVDistill	$L \rightarrow C$	R101	0.468	0.389
+DistillBEV	$L \rightarrow C$	R101	0.545	0.446
+STXD	$L \rightarrow C$	R101	0.543	0.440
+TAKD (Ours)	$L \rightarrow C$	R101	0.558	0.451
BEVDepth [22]	C	R50	0.440	0.317
+STXD	$L \rightarrow C$	R50	0.483	0.371
+DistillBEV	$L \rightarrow C$	R50	0.510	0.403
+LabelDistill [19]	$L \rightarrow C$	R50	0.528	0.419
+TAKD (Ours)	$L \rightarrow C$	R50	0.537	0.430
BEVDepth	C	R101	0.535	0.412
+DistillBEV	$L \rightarrow C$	R101	0.547	0.450
+LabelDistill	$L \rightarrow C$	R101	0.553	0.451
+TAKD (Ours)	$L \rightarrow C$	R101	0.564	0.466

• Experiments

Table 3. Compared the effect of distillation losses using various teacher, TA, and student models. * indicates the insertion of the ground truth depth. We use **bold** and underline to indicate the best and the second-best results, respectively.

Model Types			AP_{3D}			AP_{BEV}		
Teacher model	TA model	Student model	Easy	Mod.	Hard	Easy	Mod.	Hard
None	None	CaDDN	23.57	16.31	13.84	30.28	21.53	18.90
CenterPoint [48]	CaDDN*	CaDDN	27.06	19.38	17.50	35.66	26.03	23.09
PointPillar [20]	CaDDN*	CaDDN	31.28	20.80	17.58	<u>42.21</u>	<u>28.48</u>	<u>25.73</u>
Second [45]	CaDDN*	CaDDN	34.36	22.61	19.88	42.86	29.41	26.47
CenterPoint	MonoDETR*	CaDDN	26.56	18.84	16.46	34.62	25.85	22.29
PointPillar	MonoDETR*	CaDDN	28.11	20.00	17.24	40.21	27.10	24.65
Second	MonoDETR*	CaDDN	30.74	20.35	17.69	41.75	28.95	24.80
None	None	MonoDETR	28.84	20.61	16.38	37.86	26.95	22.80
CenterPoint	MonoDETR*	MonoDETR	30.78	21.17	18.41	39.73	27.22	24.69
PointPillar	MonoDETR*	MonoDETR	31.25	21.47	<u>18.57</u>	38.68	27.16	24.83
Second	MonoDETR*	MonoDETR	<u>33.18</u>	<u>21.97</u>	18.55	41.98	28.43	25.24

• Experiments

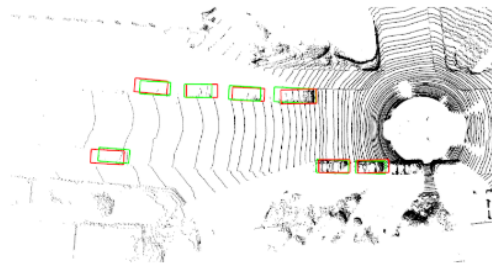
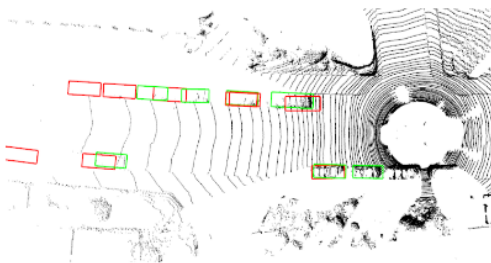
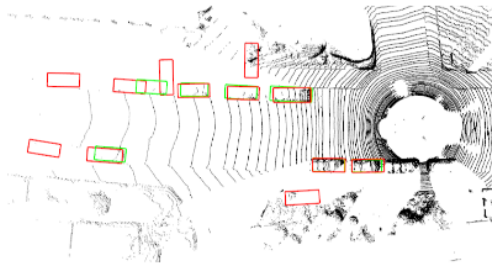
Table 4. Effectiveness of different feature distillation losses.

Settings	Loss			AP_{3D}		
	IMD	CMD	CMRD	Easy	Mod.	Hard
1				24.35	16.16	13.44
2	✓			31.11	20.24	16.91
3		✓		28.22	18.29	15.10
4			✓	29.68	19.57	16.22
5	✓	✓		30.73	19.88	16.43
6	✓		✓	34.36	22.61	19.88

Table 5. Ablation of the components of MonoTAKD. DConv. indicates the deformable convolution.

Components				AP_{3D}		
ASPP	DConv.	SENet	FFM	Easy	Mod.	Hard
				31.36	20.80	16.53
✓				32.28	21.45	17.18
	✓			32.46	21.51	17.99
✓	✓			33.19	21.81	18.23
✓	✓	✓		33.26	21.47	18.61
✓	✓	✓	✓	34.36	22.61	19.88

- Experiments - Appendix



(a) CMKD

(b) MonoDETR

(c) MonoTAKD (ours)

Figure F1. Qualitative results on KITTI *val* set for the Car category. We compare the qualitative results among CMKD [12], MonoDETR [49], and our proposed MonoTAKD. The first and second rows represent detection results from a camera frontal view and a BEV, respectively. We use green and red boxes to indicate the ground truth and prediction bounding boxes.

- Experiments - Appendix

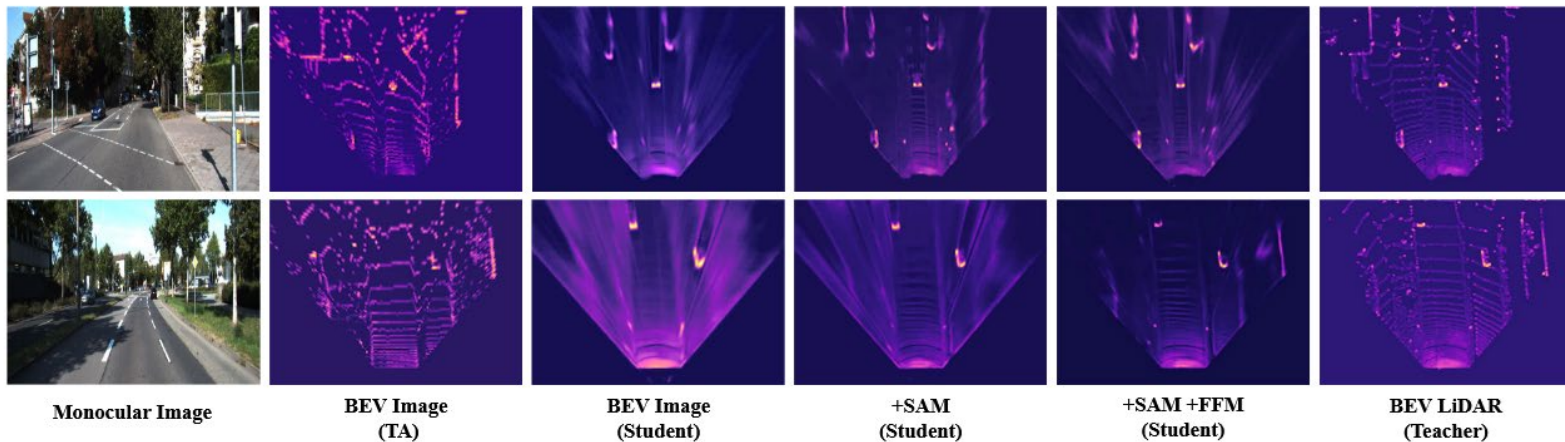


Figure F2. Visualization of the BEV features from the teacher, TA, and the two distillation branches of the student model on KITTI *val* set.